

BEN-GURION UNIVERSITY OF THE NEGEV

School of Electrical and Computer Engineering

Sparse Recovery and Compressed Sensing

A Model-Based Deep Learning Final Project

From ISTA baselines to lightweight unfolded networks

Course: Model-Based Deep Learning (361.2.2320)

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Code: https://github.com/ThEpiCake/hyperlista-mbdl

1. Motivation, background and project goals

In this project we studied sparse recovery through the model-based deep learning approach. We started from the classical ISTA and FISTA algorithms and then implemented unfolded versions with different amounts of learned flexibility. The practical question that guided the project was simple: when the network depth is fixed, which parts of ISTA are actually worth learning?

We compared methods ranging from zero learned parameters to millions of parameters. LISTA learns full update matrices, ALISTA keeps an analytic matrix and learns layer-wise scalars, and HyperLISTA uses only three tuned constants to generate its parameters adaptively. Between these existing approaches, we designed three controlled ISTA variants that learn only the thresholds, only the step sizes, or both. These variants let us isolate the contribution of each scalar sequence without changing the underlying algorithm.

1.1 What we implemented

The project contains six executed notebooks. The first establishes ISTA and FISTA baselines. The second compares LISTA, ALISTA and our simplified HyperLISTA implementation. The third contains our scalar-model study and the comparison of end-to-end, deeply supervised, greedy layer-wise and tied LISTA training. The fourth applies the models to compressed sensing of MNIST and Fashion-MNIST pixels. After the structured models performed poorly on Fashion-MNIST, the fifth notebook repeats the experiment in the DCT domain. The sixth notebook repeats the main comparisons across several random seeds.

2. Problem formulation

We observe a signal only through an underdetermined linear system. The measurement vector b is generated from a signal x^* and a known sensing matrix A:

$$b = Ax^* + \varepsilon, \quad A \in \mathbb{R}^{m \times n}, \quad m < n$$

Sparse recovery assumes that x^* has only a small number of non-zero coefficients, either in the original representation or after a transform. A common convex estimator is the LASSO:

$$\hat{x} = \arg\min_x \frac{1}{2}\|b - Ax\|_2^2 + \lambda\|x\|_1$$

ISTA solves this problem by alternating a gradient step and soft-thresholding. FISTA adds Nesterov momentum. We unroll a fixed number $K = 16$ of these iterations so that every method has the same depth budget. We report normalized MSE in dB (NMSE). For images we additionally report PSNR and SSIM.

3. Methods

3.1 ISTA, FISTA and LISTA

ISTA uses a step size $1/L$, where L is the squared spectral norm of A, and a threshold λ/L. LISTA replaces the fixed ISTA matrices with a pair of learned matrices in every layer:

$$x^{k+1} = S_{\theta_k}(W_x^k x^k + W_y^k b)$$

We initialize LISTA at the exact ISTA parameterization. Before training, the unfolded network therefore reproduces K ISTA iterations. With $m = 250$, $n = 500$ and $K = 16$, untied LISTA has 6,000,016 trainable parameters. We also test a tied version that shares one update across all layers and uses 375,001 parameters.

3.2 ALISTA

ALISTA keeps a fixed correction matrix W computed from A and learns only a step size and threshold for each layer. Its partial soft-thresholding operator leaves the p_k largest coefficients unchanged, where p_k is a fixed, non-learned schedule, and shrinks the remaining coefficients. Our 16-layer ALISTA therefore has 32 learned parameters.

$$x^{k+1} = S_{p_k,\theta_k}(x^k + \gamma_k W^T(b - Ax^k))$$

3.3 Simplified HyperLISTA variant

HyperLISTA replaces the layer-wise learned scalars with three global constants c_1, c_2 and c_3. At each layer these constants generate a residual-dependent threshold, a momentum value and a support-selection count. We find the three constants using a coarse-to-fine validation grid search, without backpropagation.

$$\theta^k = c_1\mu \left\| A^{+(Ax^k-b)} \right\|_1, \quad \beta^k = c_2\mu \|x^k\|_0$$

$$p^k = c_3 \log\left(\frac{\|A^{+b}\|_1}{\left\|A^{+(Ax^k-b)}\right\|_1}\right)$$

Here, A^+ is the Moore–Penrose pseudoinverse of A, and μ is the mutual coherence associated with $W^T A$.

Our implementation keeps these adaptive formulas, partial thresholding and a momentum term based on the previous update. For the fixed matrix, however, we use the same analytic construction as our ALISTA implementation instead of solving the original paper's D, G symmetric-Jacobian problem. We also aggregate θ and p with the batch median and omit the paper's final conjugate-gradient stage. Throughout this report, 'HyperLISTA' refers to this simplified fixed-depth variant.

3.4 Our ISTA-derived scalar models

To understand why LISTA improves over ISTA, we separated the two scalar quantities that appear in an ISTA layer: the step size and the threshold. This produced three small models. All scalar parameters are constrained to remain positive and are initialized at the ISTA values.

Table 1. Scalar models designed for the project.

Model	Learned quantities	Coupling	Parameters
ThresholdISTA	$\theta_1, \ldots, \theta_{16}$	Step fixed at $1/L$	16
StepISTA	$\gamma_1, \ldots, \gamma_{16}$	$\theta_k = \lambda\gamma_k$	16
StepThresholdISTA	γ_k and θ_k	Independent	32

ThresholdISTA asks whether changing only the shrinkage level is enough. StepISTA asks whether the finite-depth network mainly needs a better update schedule while keeping the LASSO coupling between step and threshold. StepThresholdISTA gives both sequences independent freedom. Because the models differ by only one design choice, comparing their final NMSE values shows the contribution of learning the step sizes, the thresholds, or both.

3.5 Training strategies

For learned models we use Adam, validation-based early stopping and gradient clipping. We compare three losses for untied LISTA. L1 uses only the final-layer reconstruction loss. L2 adds weighted intermediate-layer losses, which gives every layer a direct training signal. L3 trains one layer at a time while freezing earlier layers. We also train tied LISTA to test how much accuracy is lost when all layers share their parameters.

3.6 Experimental setup

Table 2. Experimental datasets and recovery settings.

Part	Signals	Measurements	Test set
Synthetic	$n = 500$, $s = 50$ Gaussian-sparse	$m = 250$, noiseless	2,048 test signals
Images	28×28 MNIST / Fashion-MNIST	ratios 0.25 and 0.50	10,000 test images

The main synthetic experiments use 51,200 training signals, 2,048 validation signals and 2,048 test signals. For each image dataset, the standard 60,000-image training set is divided into 55,000 training images and 5,000 validation images, while the standard 10,000-image test set is kept for final evaluation. Each run uses the same sensing matrix for its training, validation and test sets. The multi-seed experiment repeats the complete procedure with an independently generated sensing matrix for each seed.

4. Synthetic sparse-recovery results

The synthetic experiment matches the assumptions used by the structured methods: the signals are exactly sparse, their non-zero values are Gaussian, and the measurements are noiseless. Table 3 reports the final test NMSE after 16 layers. Lower NMSE is better.

Table 3. Synthetic test results at $K = 16$.

Method	NMSE (dB)	Parameters
ISTA	-5.3	0
FISTA	-11.0	0
ThresholdISTA (ours)	-7.1	16
StepISTA (ours)	-20.0	16
StepThresholdISTA (ours)	-23.6	32
LISTA-L1	-20.4	6,000,016
LISTA-L2	-22.2	6,000,016
LISTA-L3	-16.6	6,000,016
LISTA-Tied-L1	-19.3	375,001
ALISTA	-30.0	32
HyperLISTA	-62.9	3 tuned

The result shows that more learned parameters do not automatically improve recovery. StepThresholdISTA reaches -23.6 dB with 32 scalars, compared with -22.2 dB for the best 6-million-parameter LISTA configuration. ALISTA performs better at -30.0 dB, and HyperLISTA reaches -62.9 dB after grid-searching only three constants. These results apply to this fixed sensing matrix and matched synthetic distribution.

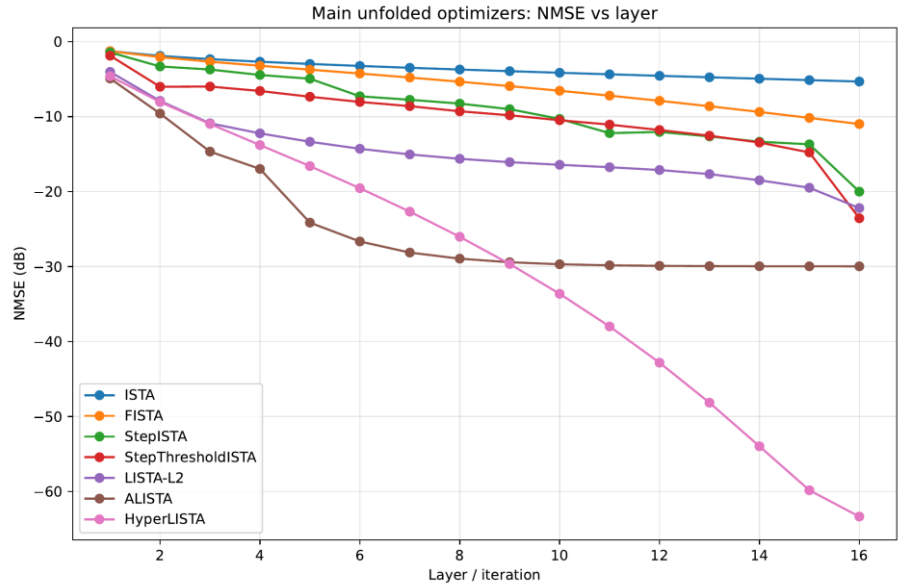


Figure 1. NMSE across the 16 unfolded layers for the main synthetic models.

4.1 What was worth learning?

The scalar-model comparison gives a direct answer to our main design question. Learning thresholds alone improves ISTA from -5.3 to -7.1 dB, which is a relatively small gain. Learning only the 16 step sizes improves the result to -20.0 dB, close to end-to-end LISTA. Allowing the steps and thresholds to vary independently gives another 3.5 dB improvement.

This behavior makes sense for a truncated algorithm: with only 16 layers, the learned step sizes can optimize progress at a fixed depth instead of following the conservative step size required by classical ISTA convergence. The threshold sequence is still useful, but most of the observed improvement comes from learning the step sizes.

4.2 Training LISTA

Deep supervision (L2) is the best LISTA training method in our experiment, reaching -22.2 dB instead of -20.4 dB for the final-loss-only method. Greedy layer-wise training stops improving after the middle layers and finishes at -16.6 dB. Tying the weights reduces the number of parameters by a factor of 16 and loses only about 1.2 dB relative to end-to-end untied LISTA.

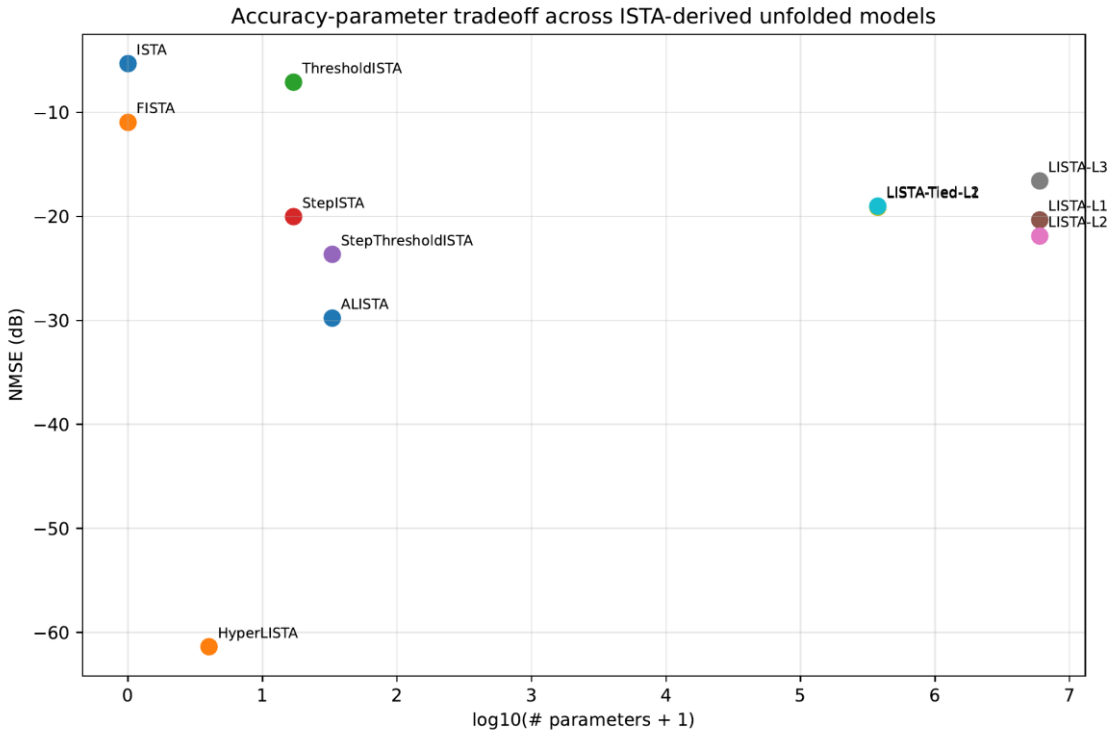


Figure 2. Accuracy versus parameter count. The scalar models occupy the useful low-parameter region.

For this synthetic problem, learning complete matrices is therefore not the most efficient use of the parameter budget. This conclusion should not be generalized to every inverse problem, because it depends on having a fixed sensing matrix and data that follow the assumed sparse model. The image experiments test what happens when that match is weaker.

5. Pixel-domain compressed sensing

We next applied the same 16-layer methods to flattened 28×28 MNIST and Fashion-MNIST images. The measurement ratio is $r = m/d$, where $d = 784$ is the number of pixels and m is the number of compressed measurements. Ratios 0.25 and 0.50 therefore correspond to $m = 196$ and $m = 392$, respectively. The measurements $b = Ax$ are random linear combinations of all pixels, so information is lost by reducing each image from 784 values to $m < 784$ measurements rather than by removing specific pixels. Before running the experiment, we expected the large number of zero pixels to make direct pixel-domain sparse recovery reasonable, especially for MNIST.

For image quality, PSNR measures pixel-level reconstruction fidelity in decibels, while SSIM measures the preservation of luminance, contrast and structural information. Higher values indicate better reconstruction for both metrics, with an SSIM of 1 indicating a perfect match.

The expectation was only partly correct. Table 4 shows the most difficult case, Fashion-MNIST at ratio 0.25. LISTA-L2 reconstructs the images well, while the classical and structured methods remain close to the zero estimate.

Table 4. Fashion-MNIST pixel-domain results at measurement ratio 0.25.

Method	NMSE (dB)	PSNR (dB)	SSIM
ISTA	-1.1	8.8	0.13
FISTA	-1.1	8.7	0.13
StepThresholdISTA	-1.1	8.8	0.14
LISTA-L2	-14.1	22.5	0.83
ALISTA	-0.8	8.4	0.13
HyperLISTA	-0.3	7.7	0.09

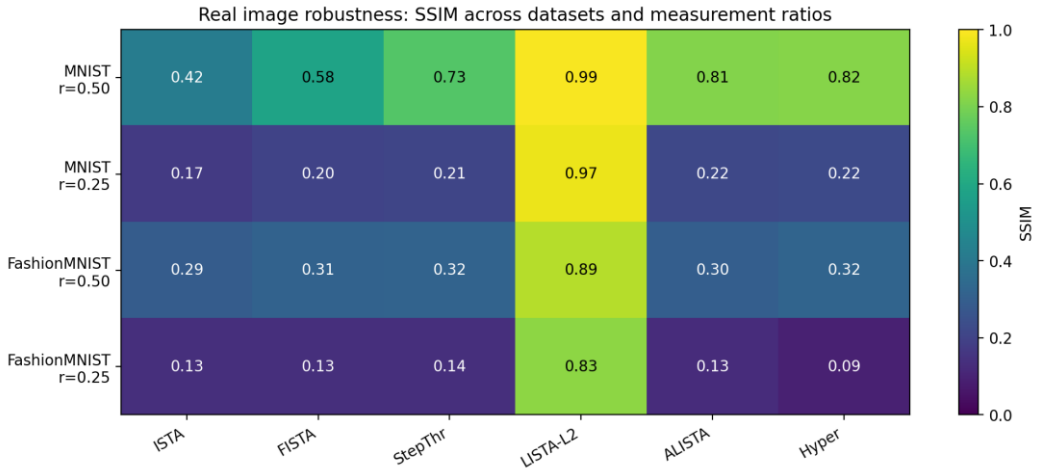


Figure 3. SSIM for both datasets and both measurement ratios in the pixel domain.

The difference can be explained by model mismatch. Fashion-MNIST pixels are non-negative, spatially correlated and only approximately sparse, so pixel-domain soft-thresholding is a poor match for the data. Although the scalar parameters and HyperLISTA constants are trained or tuned using the image training and validation sets, the structured methods retain fixed correction directions determined by the sensing matrix. LISTA learns full update matrices, giving it enough flexibility to adapt to the image distribution from the 55,000 training examples.

6. DCT-domain recovery

The poor Fashion-MNIST result raised a question: were the structured algorithms unsuitable for images, or were pixels simply the wrong sparse representation? To test this, we kept the same measurements $b = Ax$ and replaced the recovery dictionary with $D = A\Psi^T$, where Ψ is the orthonormal two-dimensional DCT. Each method recovers DCT coefficients α, and the final image is obtained with the inverse DCT.

Table 5 compares the pixel and DCT domain results for Fashion-MNIST at $r = 0.25$. Every structured method improves in the DCT domain. StepThresholdISTA and HyperLISTA are effectively tied at approximately -7.7 dB and 0.52 SSIM, despite using 32 learned scalars and only three tuned constants, respectively. Both outperform ALISTA, which reaches -5.1 dB and 0.40 SSIM. LISTA-L2 remains best, but its advantage over the strongest structured methods decreases substantially.

Table 5. Same Fashion-MNIST measurements, different recovery representation.

Method	NMSE: pixel → DCT	SSIM: pixel → DCT
ISTA	$-1.1 \rightarrow -4.0$	$0.13 \rightarrow 0.26$
FISTA	$-1.1 \rightarrow -5.6$	$0.13 \rightarrow 0.35$
StepThresholdISTA	$-1.1 \rightarrow -7.7$	$0.14 \rightarrow 0.52$
LISTA-L2	$-14.1 \rightarrow -14.9$	$0.83 \rightarrow 0.81$
ALISTA	$-0.8 \rightarrow -5.1$	$0.13 \rightarrow 0.40$
HyperLISTA	$-0.3 \rightarrow -7.7$	$0.09 \rightarrow 0.52$

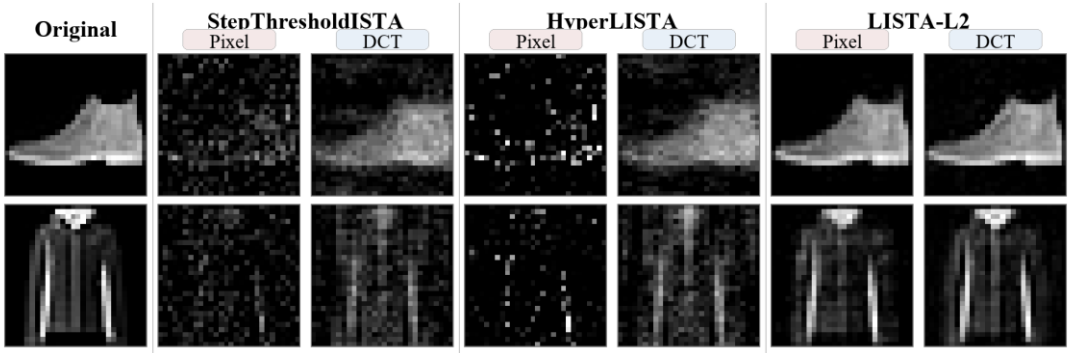


Figure 4. Representative Fashion-MNIST reconstructions at measurement ratio 0.25 using the same measurements. StepThresholdISTA and HyperLISTA largely collapse in the pixel domain but recover recognizable structure with DCT, while LISTA-L2 remains strong in both representations.

DCT is not always better. MNIST images are already highly sparse in the pixel domain because most background pixels are zero. In the corresponding MNIST experiment at $r = 0.50$, not shown in Table 5, moving to DCT reduced SSIM for the learned methods, although some obtained slightly better NMSE. Our conclusion is therefore narrower: in our experiments, structured sparse-recovery methods work best when the chosen representation provides a strong sparsity prior.

7. Discussion, robustness and limitations

Taken together, the results show that performance depends not only on the number of learned parameters, but also on how well the model structure and recovery representation match the data. Structured low-parameter methods perform best on the matched synthetic problem, while the more flexible LISTA model is substantially better for pixel-domain images. Moving Fashion-MNIST recovery to the DCT domain narrows this gap, confirming that the choice of sparse representation is an important part of the model. We next examine whether these conclusions remain stable across random seeds and then discuss the main limitations of the study.

7.1 Repeating the experiments across seeds

Most tables in this report use seed 42. To check whether the main conclusions depended on one random realization, we repeated the synthetic comparison with five seeds. For each seed, we generated a new sensing matrix and synthetic dataset and repeated model initialization, training and HyperLISTA tuning. We also repeated every pixel-domain dataset and measurement-ratio combination with three seeds, as well as the main DCT experiment on Fashion-MNIST at $r = 0.25$.

Table 6. Selected multi-seed results (mean +/- standard deviation NMSE in dB).

Experiment	Model	NMSE (dB)
Synthetic, 5 seeds	LISTA-L2	-22.37 ± 0.16
Synthetic, 5 seeds	StepThresholdISTA	-23.57 ± 0.18
Synthetic, 5 seeds	ALISTA	-29.96 ± 0.03
Synthetic, 5 seeds	HyperLISTA	-62.54 ± 0.57
Fashion DCT 0.25, 3 seeds	StepThresholdISTA	-7.62 ± 0.11
Fashion DCT 0.25, 3 seeds	HyperLISTA	-7.63 ± 0.22

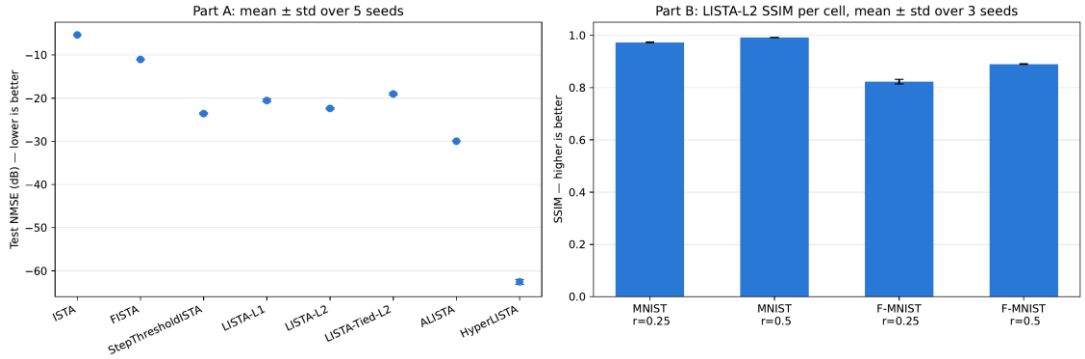


Figure 5. Multi-seed results for synthetic NMSE across models (left) and pixel-domain LISTA-L2 SSIM across datasets and measurement ratios (right).

The standard deviations are small compared with the principal performance gaps, so the main synthetic ordering is stable across the tested seeds. In the Fashion-MNIST DCT experiment, StepThresholdISTA and HyperLISTA remain effectively tied. These results do not establish robustness to noise, unseen datasets or changes in the measurement process; they show only that the main conclusions are not caused by one lucky random seed or sensing matrix.

7.2 Limitations

All experiments are noiseless, and we did not test transfer from a model trained on one sensing matrix to a different matrix at test time. HyperLISTA is a simplified fixed-depth implementation, as described in Section 3.3. Finally, image performance was evaluated only on MNIST and Fashion-MNIST. Noise experiments, cross-matrix transfer and richer transforms such as wavelets would be natural extensions.

8. Conclusions

This project compared classical sparse-recovery algorithms, unfolded learned models and lightweight structured alternatives under the same 16-layer budget. On matched synthetic data, the most structured methods were both smaller and more accurate. Our scalar-model comparison showed that learning the step-size schedule produced most of the improvement over ISTA, while learning thresholds alone provided only a limited gain. StepThresholdISTA slightly outperformed the best LISTA configuration with only 32 learned parameters.

The real-image experiments showed why these results must be interpreted together with the signal model. Direct pixel recovery worked well for data-driven LISTA but poorly for the structured methods on Fashion-MNIST. Repeating the experiment in the DCT domain recovered much of their performance, while also showing that DCT is not universally beneficial when the pixel representation is already sparse, as in MNIST. The representation is therefore part of the model, not just a preprocessing detail.

Our practical takeaway is to learn only what the known model cannot describe. When the sparse model and recovery domain match the data, a small structured network can be very effective. When those assumptions are uncertain, a more flexible trained model may be safer, although it requires many more parameters and more training data.

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